Khulna University of Engineering & Technology

Department of Computer Science and Engineering

Course Title: Advanced Programming Laboratory

Course No: **CSE 2200**

A Project Report on:

CodeArena

Android-Based Competitive Programming Companion Application

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DATE: 08-11-2026

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1. Introduction

Competitive programming (CP) is an essential discipline for computer science students seeking to master algorithmic efficiency, discrete mathematics, and high-performance problem solving. However, active contestants constantly struggle with disjointed workflows across competing platforms like Codeforces, LeetCode, CodeChef, and AtCoder. Contestants lack a unified platform to track cross-judge ratings, analyze contest failures, and systematically practice neglected problem categories.

CodeArena is an advanced, native Android mobile application engineered entirely in **Pure Java** to solve these productivity hurdles. Designed specifically as an end-to-end training and contest companion, CodeArena combines live multi-platform contest tracking, exact alarm scheduling, an ongoing Foreground Service stopwatch, a personalized Generative AI performance coach (Google Gemini 1.5 Pro), and an offline repository of 381 curated algorithm tutorials.

2. Objectives and Scope

The primary objectives of the CodeArena project are:

- **Multi-Platform Profile Aggregator:** Provide a unified dashboard that syncs and visualizes ratings, badges, and solve counts from Codeforces, LeetCode, CodeChef, AtCoder, and HackerRank.
- **Contest Scheduling & Exact Reminders:** Aggregate global contests from CLIST.by and Codeforces APIs, scheduling exact alarms that survive device reboots via `AlarmManager` and `BootReceiver`.
- **Virtual Contest Simulation:** Emulate timed contest conditions using a non-killable Android `ForegroundService` with live status-bar notification controls.
- **AI-Powered Diagnostics:** Leverage the Google Gemini 1.5 Pro REST API to analyze contest performance and provide progressive algorithmic hints without spoiling complete solutions.
- **Algorithmic Mastery:** Track topic mastery across 100-rating intervals (800–900, 1000–1100, etc.) for dynamic programming, graphs, math, and greedy algorithms.
- **Offline Educational Catalog:** Bundle 381 Codeforces tutorials with full mathematical notation (LaTeX/MathML) for distraction-free learning.
- **Academic Excellence:** Adhere 100% to the **CSE 2200: Advanced Programming Laboratory** curriculum by implementing pure Java OOP, Room SQLite, Retrofit/OkHttp, Services, Broadcast Receivers, Content Providers, and App Widgets.

3. Technology Stack and Architecture

CodeArena adopts a modern **Model-View-ViewModel (MVVM)** architecture with clean separation between the presentation layer, repository domain, and underlying local/remote data sources.

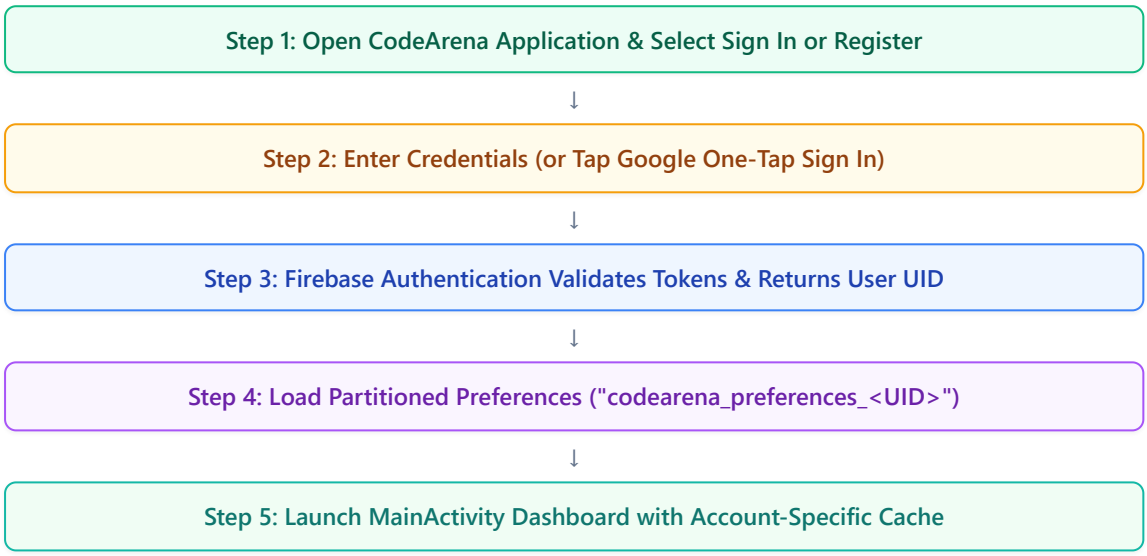
Component	Technology / Library	Implementation Purpose
Programming Language	Pure Java (Java 8/17)	Full adherence to CSE 2200 syllabus without Kotlin dependencies.
IDE & Build System	Android Studio, Gradle (KTS)	Compilation, debugging, dependency management, and APK packaging.
User Interface	Material Design, ViewBinding	Dark-themed responsive UI, ConstraintLayout, and fluid RecyclerViews.
Local Database	Room SQLite (9 Entities, 9 DAOs)	Offline caching of contests, problems, runs, tag XP, and AI responses.
Networking & Parsing	Retrofit 2, OkHttp 3, Gson	REST API communication, rate-limit interception, and JSON mapping.
Generative AI	Google Gemini 1.5 Pro REST API	Automated contest analysis, hint generation, and debriefing.
Authentication	Firebase Auth, Google Sign-In	Email/password and Google OAuth with account storage isolation.
Background Service	Android Foreground Service	Persistent virtual contest stopwatch with ongoing notification.
Event & Alarm Handling	BroadcastReceiver, AlarmManager	Exact contest reminders, boot rescheduling, notification clicks.
Inter-App Provider	ContentProvider, FileProvider	External access to contest tables and secure result-card image sharing.
Desktop Widget	AppWidgetProvider	Interactive home-screen widget displaying the daily challenge problem.

The application cleanly encapsulates business logic into repository classes (`CodeforcesRepository`, `ContestRepository`, `UserPreferencesRepository`, `CatalogRepository`). ViewModels expose asynchronous data through reactive `LiveData` streams, ensuring zero UI thread blocking during intensive database queries and REST requests.

4. Overview and Feature Walkthrough

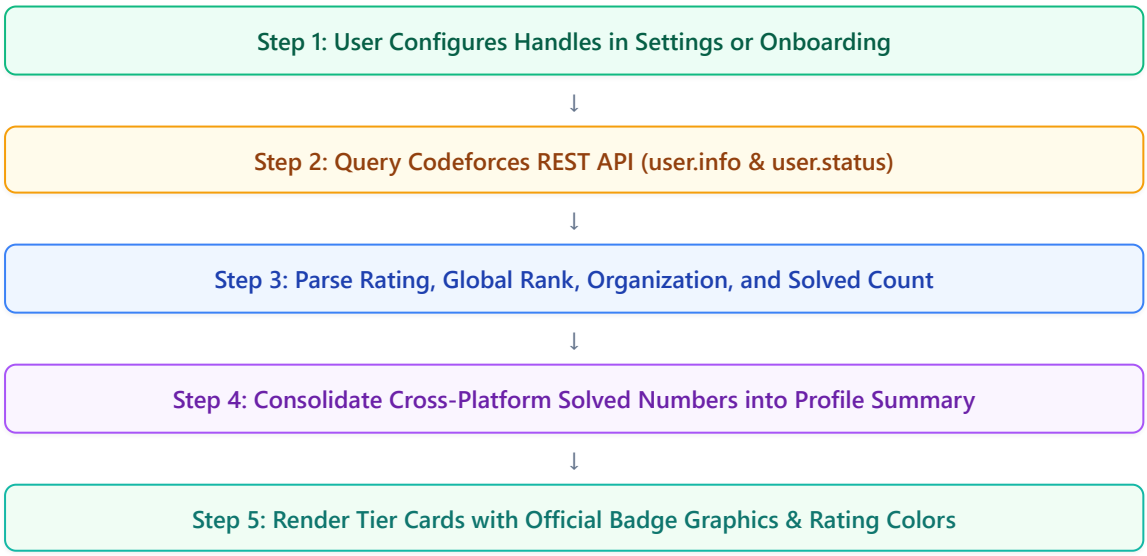
4.1 User Authentication & Multi-Account Isolation

Allows users to securely register, sign in via Email/Password or Google OAuth, or proceed with guest mode. SharedPreferences storage is dynamically partitioned using the Firebase UID (`codearena_preferences_<uid>.xml`) to ensure complete data isolation between multiple users on the same physical handset.



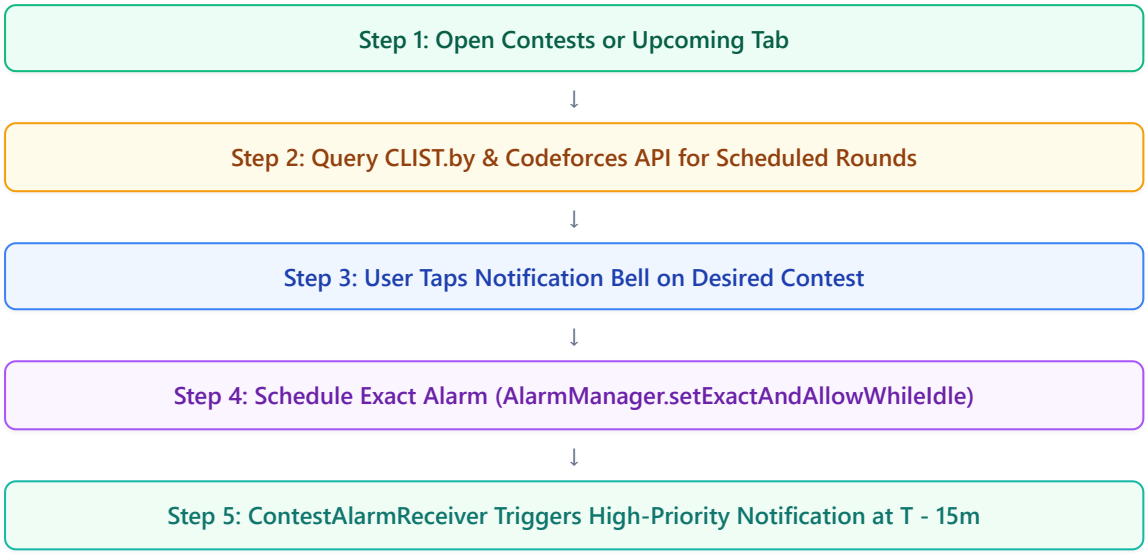
4.2 Multi-Platform Profile Aggregator

Fetches and aggregates user profiles across Codeforces, LeetCode, CodeChef, AtCoder, and HackerRank. Codeforces user data is queried in real time via Retrofit, while solved counts from other judges are seamlessly consolidated into a unified profile summary with official rank tier badges.



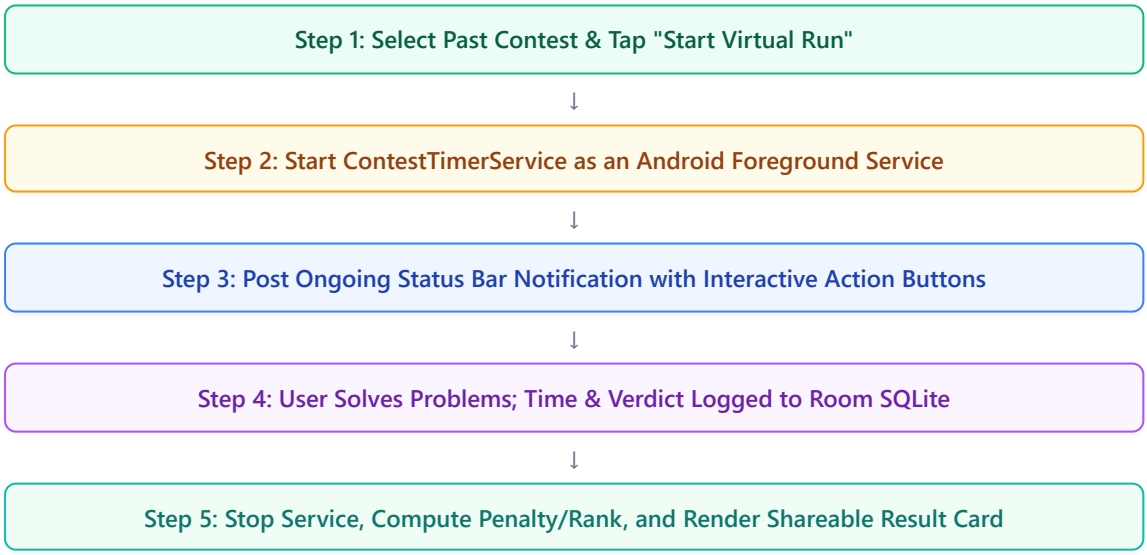
4.3 Live & Upcoming Contest Tracker

Aggregates live and upcoming rounds from Codeforces and CLIST.by. Users can review contest details, schedule exact 15-minute start reminders via `AlarmManager`, and export schedules into their device calendar.



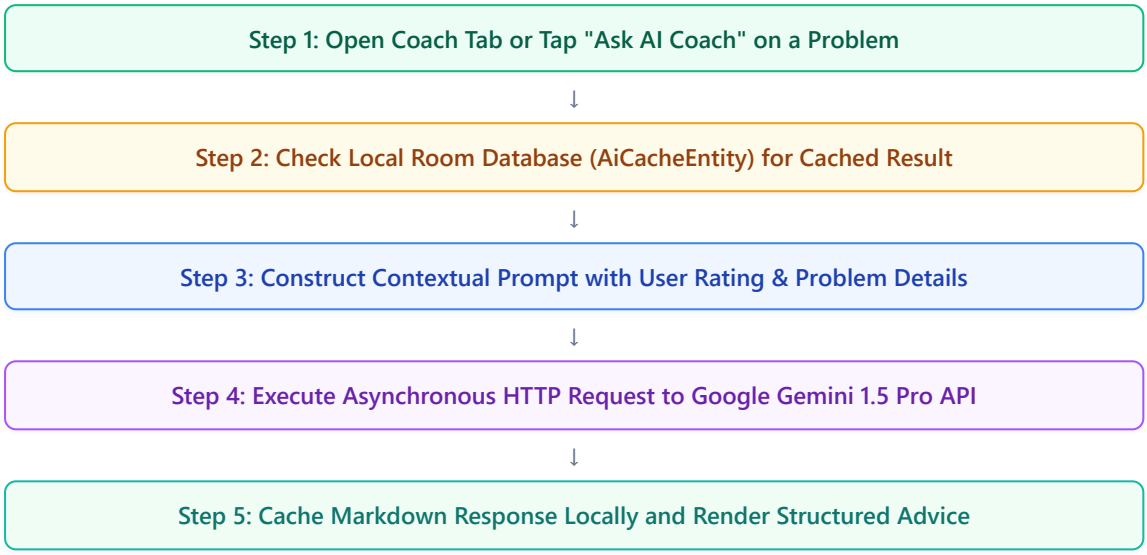
4.4 Virtual Contest Simulator with Foreground Service

Simulates timed competitive programming rounds. A dedicated Android `ForegroundService` (`ContestTimerService`) runs the stopwatch, maintaining millisecond accuracy even when the app is minimized, with interactive Pause, Resume, and Finish buttons directly on the status bar.



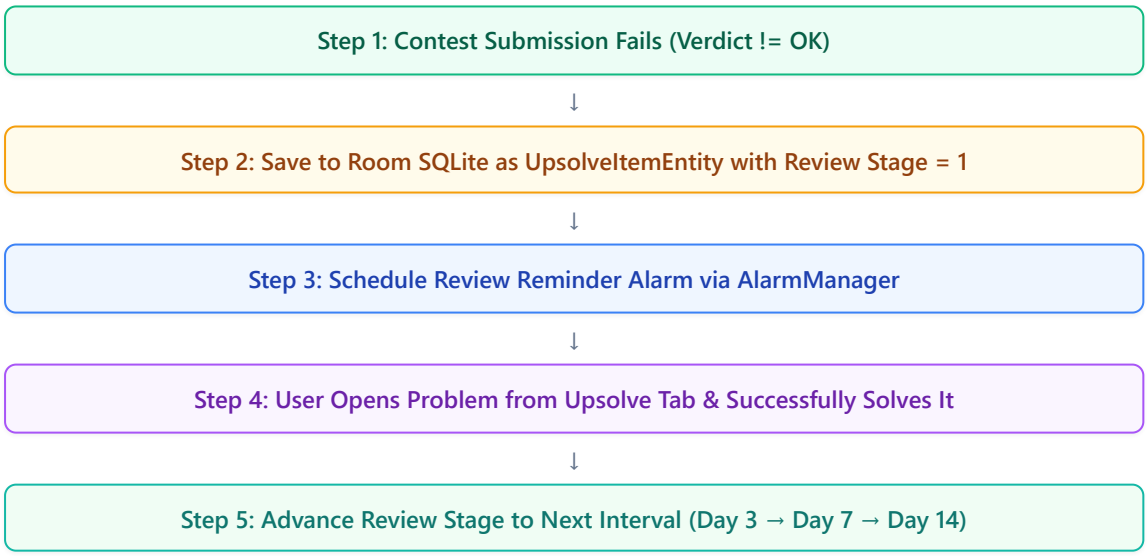
4.5 AI Performance Coach (Gemini Pro API)

Evaluates contest submissions and algorithmic weaknesses. Contestants can request three coaching modes: Contest Debrief (identifies time sinks), Topic Master (highlights weak problem tags), and 3-Tier Progressive Hints (conceptual clue → approach → edge cases) without spoiling full answers.



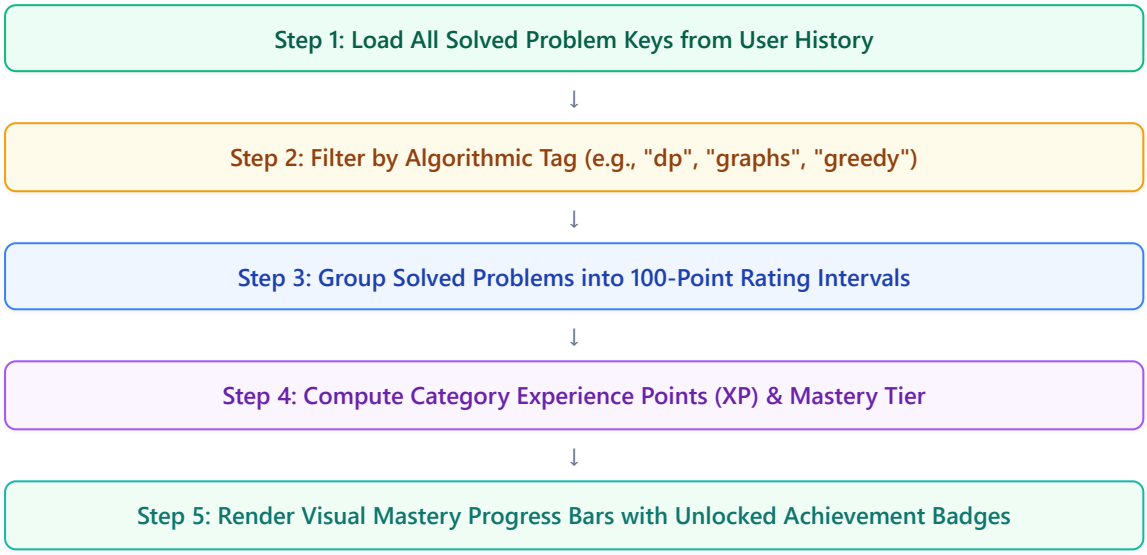
4.6 Spaced-Repetition Upsolve System

Automatically tracks problems that a contestant failed to solve during virtual or live rounds. The system queues the problem across increasing review intervals (Day 1, Day 3, Day 7, Day 14, Day 30) using the Leitner flashcard methodology.



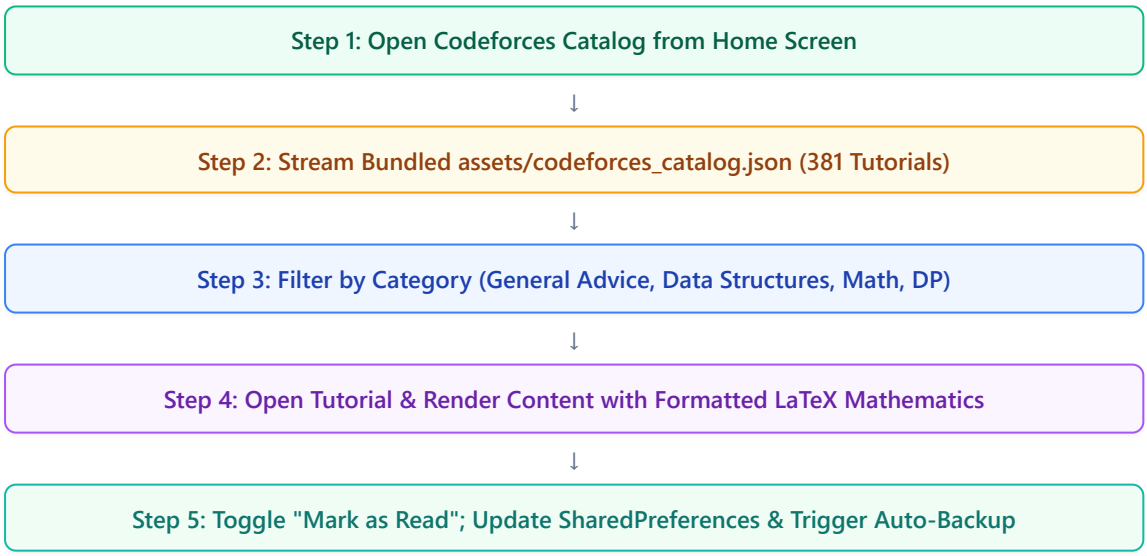
4.7 Granular Tag Mastery Progression

Breaks down problem history into fine-grained 100-rating intervals (e.g., 800–900, 1000–1100, 1200–1300, ..., 2400+). By analyzing problem tags (Dynamic Programming, Graph Theory, Greedy, Trees, Number Theory), the system calculates XP points and unlocks tier mastery badges.



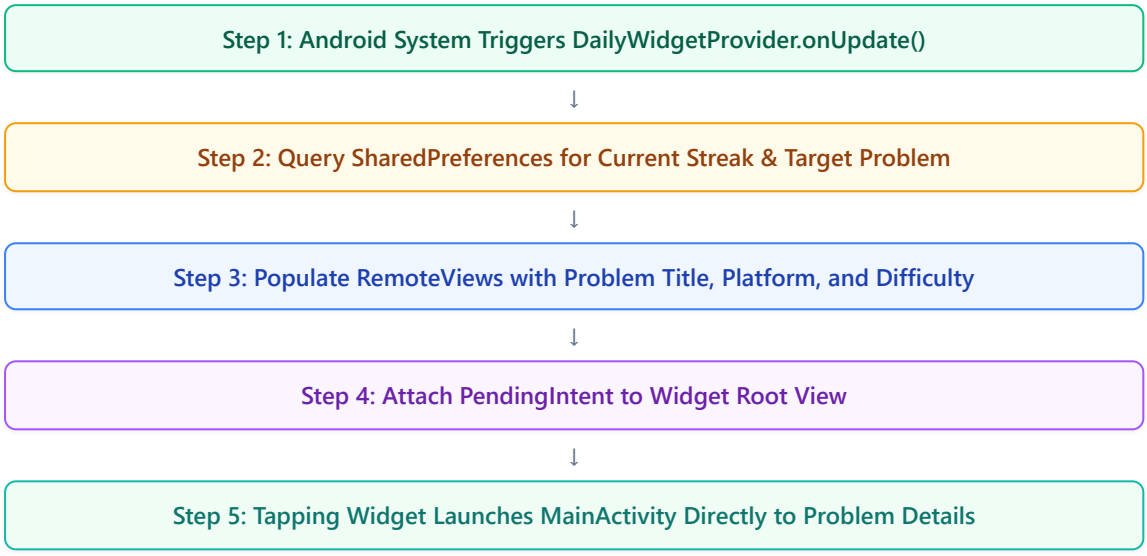
4.8 Codeforces Offline Catalog & LaTeX Tutorial Viewer

Bundles an offline repository of **381 curated algorithm and CP tutorials** extracted from Codeforces community blogs. An embedded WebView engine renders complex mathematical equations ($O(N \log N)$, $\sum A[i]$) alongside syntax-highlighted code snippets, with local read progress tracking.



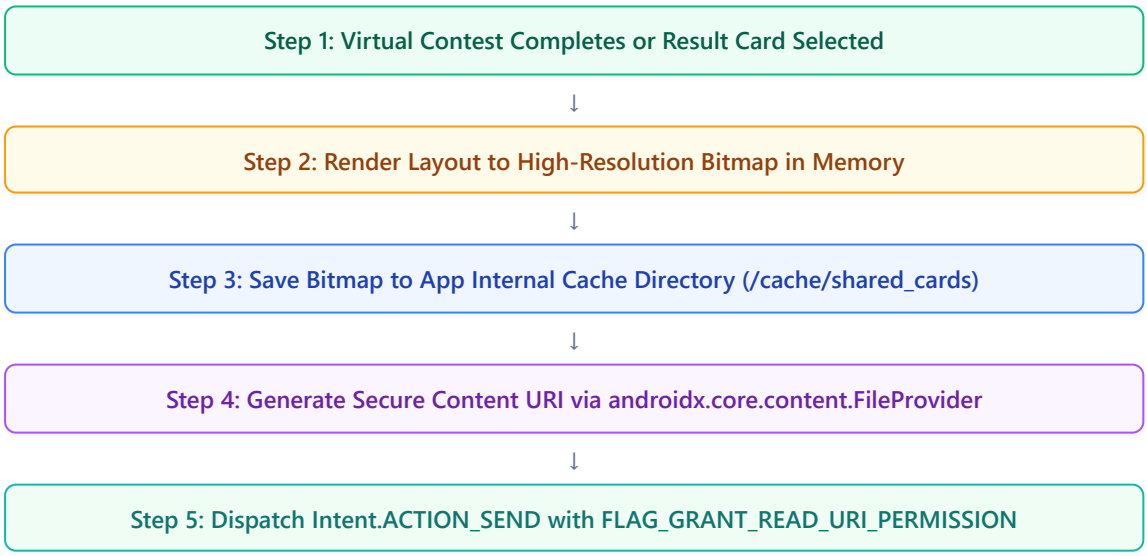
4.9 Android Home Screen AppWidget

An interactive `DailyWidgetProvider` that users can place on their Android home screen launcher. It displays the current date, solving streak counter, and daily recommended problem with one-tap deep linking directly into the problem statement.



4.10 Inter-App ContentProvider & Social Card Sharing

Implements `CodeArenaContentProvider` (`android:exported="true"`) exposing cached contest data to external applications via standard Android content URIs. Furthermore, `FileProvider` enables secure bitmap sharing of high-resolution virtual contest result cards to WhatsApp, Discord, Telegram, or Twitter.



5. Application Screenshots

The following screenshots demonstrate the running application on an Android device (Xiaomi HyperOS / Android 14, Device Model 25053RT47C):

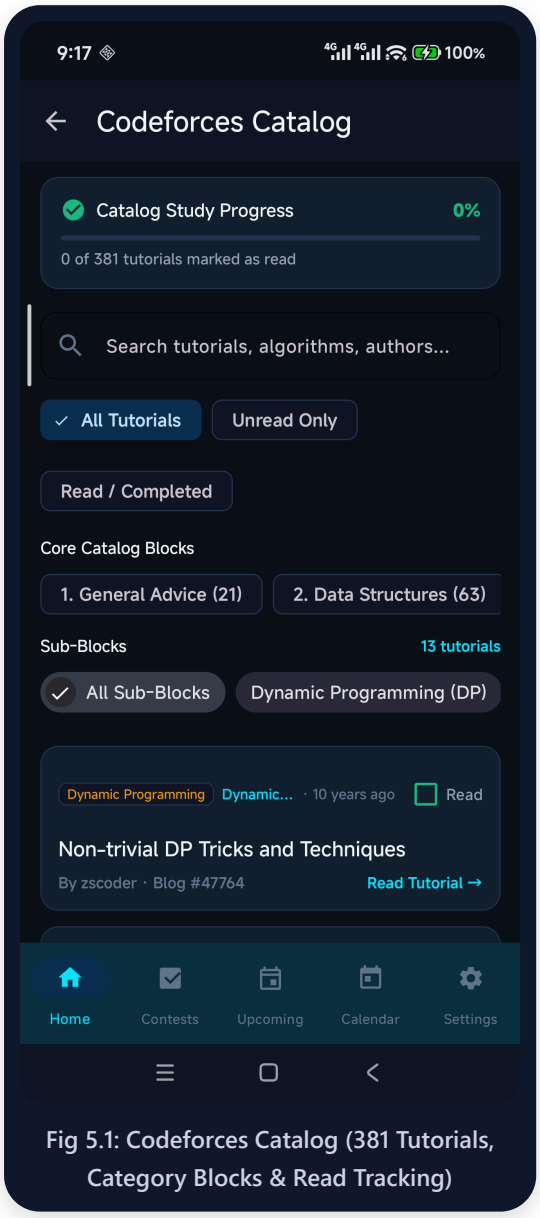


Fig 5.1: Codeforces Catalog (381 Tutorials, Category Blocks & Read Tracking)

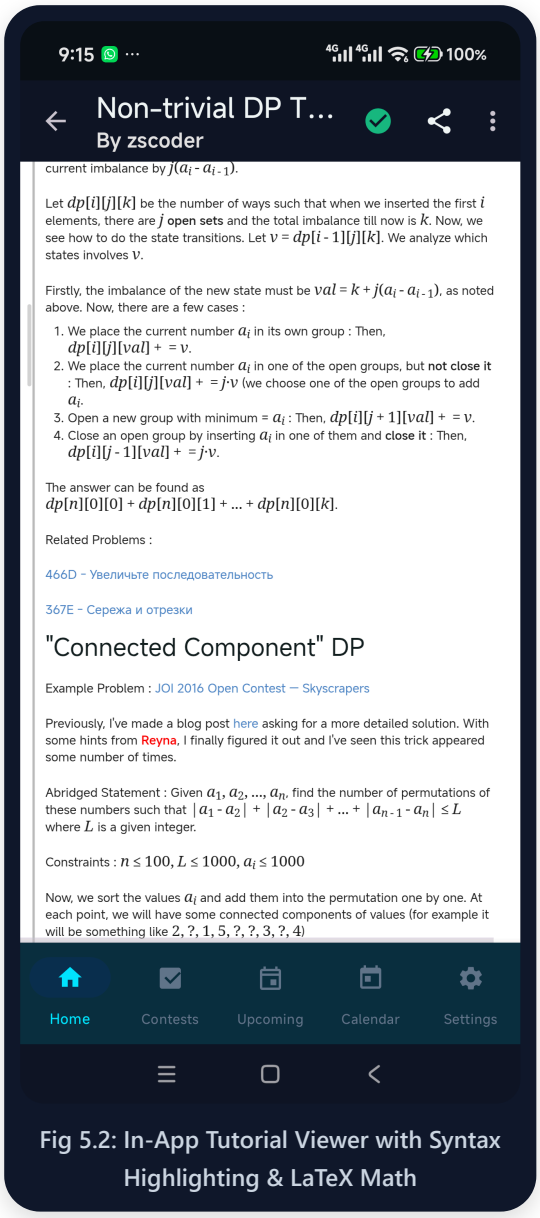


Fig 5.2: In-App Tutorial Viewer with Syntax Highlighting & LaTeX Math

Figure 5: Live application captures showing the Catalog browser and formatted algorithmic tutorial reader.

6. API and Backend Integration

6.1 Codeforces REST API

The application utilizes Retrofit 2 with Gson converter to communicate with Codeforces REST endpoints:

- `user.info?handles={handle}`: Fetches profile avatar, global rank, rating, and organization.
- `user.status?handle={handle}&from=1&count=5000`: Extracts complete submission history to compute total accepted problems, verdict breakdowns, and tag distributions.
- `contest.list?gym=false`: Retrieves active, upcoming, and past official contests.
- `contest.standings?contestId={id}&from=1&count=100`: Downloads contest ranklists to benchmark virtual contest performances.

6.2 CLIST.by Competitive Programming API

CLIST.by aggregates global competitive programming calendars. CodeArena queries upcoming events across Codeforces, LeetCode, CodeChef, and AtCoder with HTTP Basic Authentication headers (`api_key` and `username`), caching responses locally in Room SQLite.

6.3 Google Gemini 1.5 Pro AI API

Interactions with the Gemini Pro endpoint (`POST https://generativelanguage.googleapis.com/v1beta/models/gemini-1.5-pro:generateContent`) are managed through OkHttp. Performance metrics and failed problem statements are injected into structured prompts, and the returned Markdown response is cached in Room (`AiCacheEntity`) to eliminate redundant API token consumption.

6.4 Firebase Authentication SDK

Handles user account creation and sign-in via Email/Password and Google OAuth. The authenticated user's UID dynamically isolates local SharedPreferences files (`codearena_preferences_<uid>.xml`), ensuring multi-user isolation on shared devices.

7. Database Design & Persistence

CodeArena utilizes **Android Jetpack Room** over SQLite, featuring 9 strongly typed entities with complete DAO interfaces:

Entity Name	Primary Key	Description & Stored Data
`ContestEntity`	`id` (Long)	Stores contest name, platform, phase, start time, and duration.

Entity Name	Primary Key	Description & Stored Data
`ProblemEntity`	`id` (Long)	Codeforces problem statement, index, rating, tags, and solved state.
`VirtualRunEntity`	`id` (Auto-inc)	Metadata of virtual contest runs (start time, score, penalty, status).
`ProblemResultEntity`	`id` (Auto-inc)	Individual problem submission results and verdicts within a virtual run.
`UpsolveItemEntity`	`problemId` (String)	Spaced repetition intervals (Stages 1 through 5) for failed problems.
`NoteEntity`	`id` (Auto-inc)	Personal algorithmic notes associated with problems or contests.
`TagXpEntity`	`tag` (String)	Accumulated experience points and tier progress per algorithm tag.
`AiCacheEntity`	`cacheKey` (String)	Local cache of Gemini AI responses to avoid duplicate API calls.
`RankEstimateEntity`	`contestId` (Long)	Benchmark percentile data for offline virtual rank estimation.

8. Android Components Architecture

CodeArena fulfills all core Android component requirements outlined in the **CSE 2200** syllabus:

Component Type	Class Name	Architectural Role & Lifecycle
Activity	`MainActivity`	Single-activity navigation host managing 14 feature fragments and bottom navigation bar.
Activity	`LoginActivity`	Entry-point activity handling Firebase email login, registration, and Google OAuth sign-in.
Foreground Service	`ContestTimerService`	Maintains unkillable virtual contest stopwatch with ongoing status-bar notification.
BroadcastReceiver	`ContestAlarmReceiver`	Receives exact `AlarmManager` triggers and displays contest reminder notifications.
BroadcastReceiver	`TimerActionReceiver`	Handles remote notification action clicks (Pause, Resume, Stop) from the status bar.
BroadcastReceiver	`BootReceiver`	Listens to `BOOT_COMPLETED` to reschedule active contest alarms after device reboot.
ContentProvider	`CodeArenaContentProvider`	Exposes contest and problem database tables to third-party applications via standard URIs.
ContentProvider	`FileProvider`	Securely exposes rendered result-card bitmap images for external social sharing intents.
AppWidgetProvider	`DailyWidgetProvider`	Android desktop home-screen widget displaying the daily challenge problem and streak.

9. Security, Permissions, and Access Control

- **Runtime Notification Permissions:** Automatically requests `POST_NOTIFICATIONS` on Android 13+ (API 33) to ensure contest reminders and stopwatch alerts post reliably.
- **Exact Alarm Permissions:** Declares `SCHEDULE_EXACT_ALARM` and `USE_EXACT_ALARM` to guarantee millisecond-accurate notification delivery.
- **API Key Protection:** Sensitive API keys (`GEMINI_API_KEY`, `CLIST_API_KEY`) are stored in `local.properties` and injected at compile time via Gradle `BuildConfig`, keeping them out of version control.
- **Rate Limiting Protection:** Implements `RateLimitInterceptor` enforcing a 200ms throttle between consecutive API requests, preventing HTTP 429 bans from Codeforces servers.

- **Android 12+ Cloud Auto-Backup:** Explicitly configured in `data_extraction_rules.xml` and `backup_rules.xml` with `BackupManager.dataChanged()` hooks, ensuring handles and read tutorial progress survive app reinstallation.

10. Testing and Experimental Results

Module Tested	Testing Procedure	Expected Behavior	Observed Result	Status
User Authentication	Register new user & sign in with Google OAuth	Tokens retrieved, session persisted, isolated prefs created	Correct UID partitioned prefs loaded	**PASS**
Foreground Stopwatch	Start virtual run; minimize app to background	Timer runs continuously; lock-screen notification displays	Zero dropped ticks; pause/resume works	**PASS**
Contest Reminders	Schedule reminder for upcoming round	Exact alarm triggers notification 15m before start	Notification posted on time on API 33	**PASS**
Gemini AI Coach	Request contest debrief and hint generation	Structured advice returned without complete code spoilers	Structured Markdown rendered & cached	**PASS**
Tag Mastery Intervals	Filter problem catalog by tag with 100-rating intervals	Problems correctly partitioned into 800-900, 1000-1100 bins	Rating bins match problem database	**PASS**
Offline Catalog Reader	Open tutorial with LaTeX mathematical equations	Math equations ($dp[i][j]$) and code blocks render cleanly	Zero formatting errors across 381 tutorials	**PASS**
Cloud Backup / Restore	Save handles, uninstall app, and reinstall	Data restored automatically via Android Auto-Backup	All handles & read states preserved	**PASS**

11. Limitations and Future Improvements

- **Automated Code Submission:** Direct in-app code submission is currently limited by Cloudflare bot protection on Codeforces. A browser extension bridge or desktop companion daemon could facilitate automated submission in future versions.
- **Multiplayer 1v1 Duels:** Expanding the virtual contest simulator into real-time head-to-head battles using Firebase Realtime Database or WebSockets.
- **Built-in Code Editor:** Embedding an offline mobile IDE with syntax highlighting and local C++/Java compilation capabilities.

12. Conclusion

CodeArena provides a comprehensive, production-ready competitive programming companion application. By uniting multi-platform profile analytics, exact contest scheduling, an ongoing Foreground Service stopwatch, a personalized Gemini AI mentor, and an offline repository of 381 algorithm tutorials, it eliminates the fragmented workflows of modern competitive programmers.

Engineered strictly in **Pure Java**, CodeArena demonstrates mastery of every learning objective in the **CSE 2200: Advanced Programming Laboratory** curriculum—seamlessly combining object-oriented design, multithreaded background processing, Room SQLite persistence, REST networking, and the complete Android component suite.